Into the rabbit hole: Computing HD isogenies through graph theory

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Current state of isogenies

- HD isogenies are everywhere:
  - dim 2: SQIsign (& var.), PRISM, POKE (& var.), DeuringVRF, ...
  - dim 4: SQIsignHD, (qt-)Pegasis, $\otimes$-MIKE, ...
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- Big limiting factor of isogeny based cryptography in near futur !

History rhymes

> "The security of cryptosystems based on elliptic curves is not well understood, due in large part to the abstruse nature of elliptic curves. Few cryptographers understand elliptic curves [...] Over time, this may change, but for now trying to get an evaluation of the security of an elliptic-curve cryptosystem is a bit like trying to get an evaluation of some recently discovered Chaldean poetry."
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[a] Focus on 2-isogenies.

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1. Introduce simplified theory to HD isogenies[a] !
2. Use it to gain speed-up and flexibility !
3. Explore the rabbit hole of new possibilities.

[a] Focus on 2-isogenies.

On the shoulders of giants

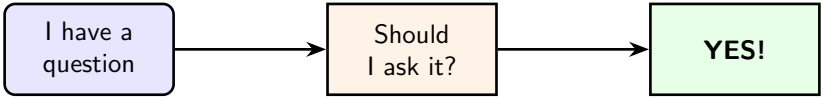
Talk based on the following works:

- [Dup25] *Superglue: Fast formulae for (2,2)-gluing isogenies*, single author, ASIACRYPT 2025.
- [DD26] *Chasing Rabbits Through Hypercubes: Better algorithms for higher dimensional 2-isogeny computations*, with Pierrick Dartois, preprint, 2026.
- $\otimes$-*MIKE for mere mortals (working title)*, with the MIKE-team *(in preparation)*.

Part is a palimpsest of the following work:

- [Dar25] *Fast computation of higher dimensional isogenies for cryptographic applications*, by Pierrick Dartois, PhD Thesis, 2025

This is an IOP



Always remember [Sha92]

$$\mathrm{IP} = \mathrm{PSPACE}$$

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   - Special isogenies

Abelian varieties

$$E$$

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- $E[N] \cong \mathbb{Z}_N^2$

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- Surjective morphism $\implies |\ker(f)| < \infty$.
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- dual $\widetilde{f} : (\overline{E}, e^{\overline{E}}) \to (E, e^E)$ with $\ker(\widetilde{f}) = f\left(E[\deg(f)]\right)$.

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- Compatible with group structures.
- $\ker(f)$ max. isotropic ($e^A|_{\ker(f)^2} \equiv 1$).
- $e^B(f(P), f(Q)) = e^A(P, Q)^{\deg(f)}$
- dual $\widetilde{f} : (B, e^B) \to (A, e^A)$ with $\ker(\widetilde{f}) = f(A[\deg(f)])$.

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- Equivalent to **symplectic basis** $\mathcal{B}_\mu := \{S_1, \ldots, S_g, T_1, \ldots, T_g\}$:

$$e_A(S_i, S_j) = e_A(T_i, T_j) = 1, \quad e_A(S_i, T_j) = \zeta^{\delta_{i,j}}$$

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satisfying the *theta group action relation*:

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- How do we construct a theta structure ?

Constructing theta structures

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Question: Can you spot the problem ?

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- $\mathfrak{t}_{T_j}$ are projective, so addition is not well defined !

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[Mum66] Mumford theorem (simplified)

Symplectic basis of $A[4] \implies \Theta^A$ theta structures (of level 2)

Example: Montgomery curves

- On Kummer line $E_{/\pm 1}$:

$$\mathfrak{g}_P \cdot \mathfrak{g}_Q = \pm \mathbf{i} \mathfrak{g}_{P+Q} = -\mathfrak{g}_Q \cdot \mathfrak{g}_P$$

[1] $u = a^2 + b^2$, $v = a^2 - b^2$

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Pauli Matrices

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad Y = \begin{pmatrix} 0 & -\mathbf{i} \\ \mathbf{i} & 0 \end{pmatrix} \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

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$$
\begin{tikzcd}
 & & (1:1) \arrow[dr] & & (1:-1) \arrow[dl] & & \\
(a+b:a-b) \arrow[dr] & & & (0:1) \arrow[d] & & & (a+\mathbf{i}b:a-\mathbf{i}b) \arrow[dl] \\
 & (u:v) \arrow[rr] & & (1:0) & & (v:u) \arrow[ll] & \\
(a-b:a+b) \arrow[ur] & & & & & & (a-\mathbf{i}b:a+\mathbf{i}b) \arrow[ul]
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$$

Figure: Structure of $E[4]_{/\pm 1}$ on Montgomery curves[1]

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 & & (1:1) \arrow[dr] & & (1:-1) \arrow[dl] & & \\
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 & (u:v) \arrow[rr] & (1:0) & & & (v:u) \arrow[ll] & \\
(a-b:a+b) \arrow[ur] & & & & & & (a-\mathbf{i}b:a+\mathbf{i}b) \arrow[ul]
\end{tikzcd}$$

Figure: Structure of $E[4]_{/\pm 1}$ on Montgomery curves[1]

Symmetric elements on Montgomery curves

$$\mathfrak{g}_{(1:\pm 1)} = \pm X \qquad \mathfrak{g}_{(a\pm b:a\mp b)} = \pm \frac{1}{2ab}\left(uZ - \mathbf{i}vY\right) \qquad \mathfrak{g}_{(a\pm \mathbf{i}b:a\mp \mathbf{i}b)} = \mp \frac{1}{2ab}\left(\mathbf{i}vZ + uY\right)$$

[1] $u = a^2 + b^2$, $v = a^2 - b^2$

Example: Theta model

Translation in the theta model

$$\mathfrak{g}_P \iff \text{theta action over } \Theta^A$$

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$$\mathfrak{g}_{S_1} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \qquad \mathfrak{g}_{S_2} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

$$\mathfrak{g}_{T_1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \quad \mathfrak{g}_{T_2} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

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Theta model $\iff$ **Fixing canonical form of** $\mathfrak{g}_P$

Duality on theta structures

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$$A[4] = \{S_1, \ldots, S_g, T_1, \ldots, T_g\}$$

$$A[4] = \{T_1, \ldots, T_g, -S_1, \ldots, -S_g\}$$

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$$A[4] = \{S_1, \ldots, S_g, T_1, \ldots, T_g\} \xleftrightarrow{\begin{pmatrix} 0 & -I_g \\ I_g & 0 \end{pmatrix}} A[4] = \{T_1, \ldots, T_g, -S_1, \ldots, -S_g\}$$

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$$\Downarrow \qquad\qquad\qquad \Downarrow$$

$$\theta^A : A \longrightarrow \mathbb{P}^{2^{g-1}} \qquad\qquad \widetilde{\theta}^A : A \longrightarrow \mathbb{P}^{2^{g-1}}$$

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$$\mathcal{H} : (x_i)_{i \in \mathbb{Z}_2^g} \longrightarrow \left(\sum_{i \in \mathbb{Z}_2^g} (-1)^{\langle j | i \rangle} x_i \right)_{j \in \mathbb{Z}_2^g}$$

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Notation:

$$(x_i)_i \odot (y_i)_i = (x_i y_i)_i \qquad\qquad (x_i)_i^{\odot n} = (x_i^n)_i \text{ for } n \in \mathbb{Z}.$$

Isogeny Formula

$$(A, e_A) \xrightarrow{f \text{ of degree } 2} (B, e_B)$$

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$$A[4] = [2](K_S \oplus K_T) \xleftarrow{\ker(\widehat{f})=[4]f(K_S)} B[4] = [2]f(K_S) \oplus f(K_T)$$

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Isogeny Formula

$$\mathcal{H}\left(\theta^A(P+Q) \odot \theta^A(P-Q)\right) = \widetilde{\theta}^B(f(P)) \odot \widetilde{\theta}^B(f(Q))$$

Arithmetic on theta coordinates

Addition Formula

$$\mathcal{H}\left(\theta^A(P+Q)\odot\theta^A(P-Q)\right)\odot\mathcal{H}\left(\theta^A(0)^{\odot 2}\right)=\left(\widetilde{\theta}^B(f(P))\odot\widetilde{\theta}^B(f(Q))\right)\odot\left(\widetilde{\theta}^B(0)^{\odot 2}\right)$$

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- [Gau07] This is a differential arithmetic on A.
  - simple !
  - fast !

Table of Contents

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   - Symmetric elements
   - Isogeny Formula
2. Computing isogenies using graphs
   - Isogeny computing
   - HIIP
3. Into the rabbit hole
   - Gluing
   - Special isogenies

Computing Isogenies

$$\prod^{g} E_i = A_0 \xrightarrow{\quad f \quad} A_e$$

Computing Isogenies

$$\begin{tikzcd}
\prod^{g} E_i = A_0 \arrow[r, "f_1"] \arrow[rrrrrrr, bend left, "f"] & A_1 \arrow[r, "f_2"] & A_2 \arrow[r, "f_3"] & \cdots \arrow[r, "f_{e-2}"] & A_{e-2} \arrow[r, "f_{e-1}"] & A_{e-1} \arrow[r, "f_e"] & A_e
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- Bottleneck: How do we compute $\widetilde{\theta}^B(0_B)^{\odot -1}$?
  - Not trivial !

Computing the inverse dual theta null point

$$\widetilde{\theta}^B(0)^{\odot -1} \quad \leftrightsquigarrow \quad \theta^B(0)$$

Computing the inverse dual theta null point

$$A[8] = \langle S_1, \cdots, S_g, T_1, \cdots, T_g \rangle$$

$$\begin{array}{ccc} A[4] = [2](K_S \oplus K_T) & \xrightarrow{\ker(f)=[4]K_T} & B[4] = [2]f(K_S) \oplus f(K_T) \\ \Downarrow & & \Downarrow \\ \Theta^A & & \Theta^B \end{array}$$

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<u>Goal:</u> [DMPR24, Dar24] Solve the equation systems

$$\left\{ \widetilde{\theta}^B_i(f(T_{\mathbf{k}})) \cdot \widetilde{\theta}^B_i(0) = \alpha_{i,\mathbf{k}} \,\middle|\, \mathbf{k} \in S \subseteq \mathbb{Z}_2^g, i \in [2^g] \right\}$$

IIP

Let:

- $S \subseteq \mathbb{Z}_2^g$ be the *support*.
- $(\pi_{i,j})_{i \in \mathbb{Z}_2^g, j \in S}$ be such:

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$$(\pi_{i,j})_{i \in \mathbb{Z}_2^g, j \in S} \xrightarrow{Compute^a} \mathbf{x}^{\odot -1} := (x_i^{-1})_{i \in \mathbb{Z}_2^g}$$

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[DD26] IIP theorem (informal)

It's just graph topology !

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Hypercube Inverse Interpolation Problem

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[DD26] HIIP theorem (informal)

It's just graph topology !

HIIP as Graph Theory

$\pi_{i,j} \leftrightsquigarrow G := (V, E)$ codomain graph

- x_i are vertices.

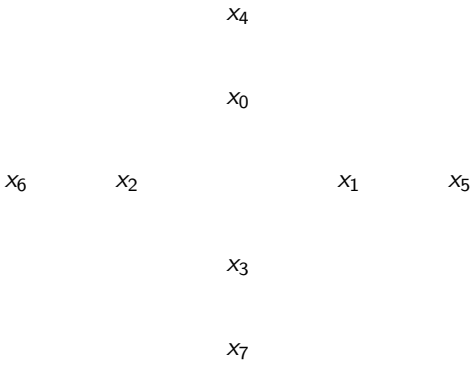

Figure: Codomain Graph G for $g = 3$, $S = \{ \quad \}$

[1] assuming $\pi_{i,i+k}, \pi_{i,i+k} \neq 0$

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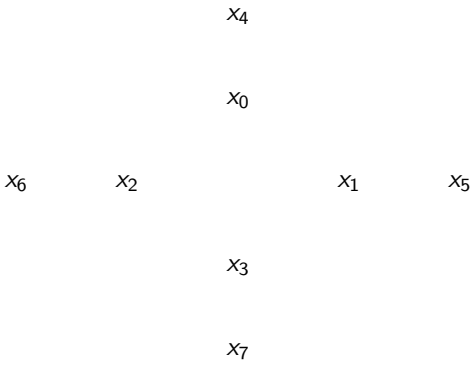

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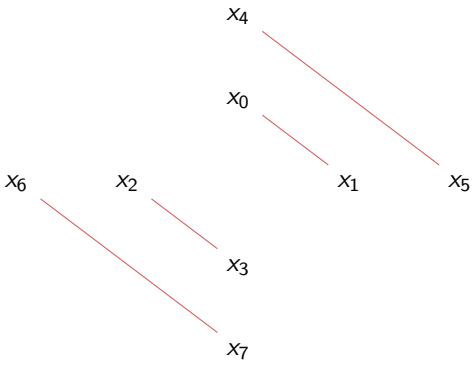

Figure: Codomain Graph G for $g = 3$, $S = \{1 \quad \}$

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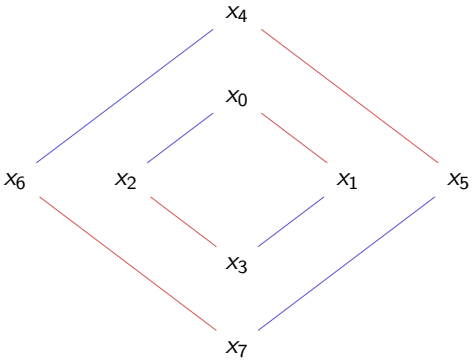

Figure: Codomain Graph G for $g = 3$, $S = \{1, 2\}$

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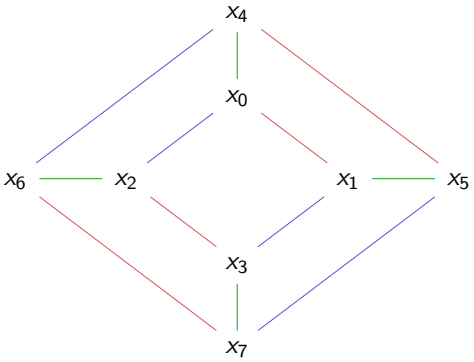

Figure: Codomain Graph G for $g = 3$, $S = \{1, 2, 4\}$

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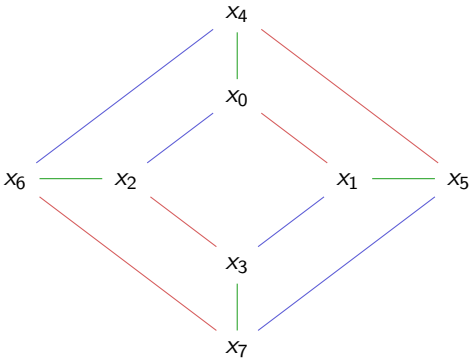
HIIP Theorem I

HIIP has a unique solution

$\iff$ G is connected



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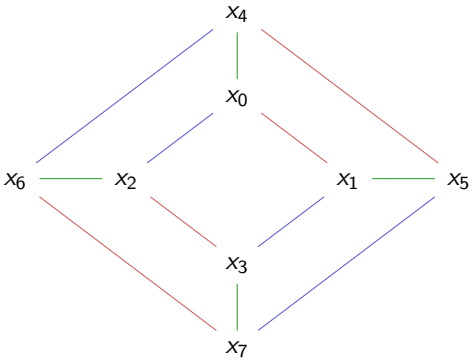
- We know when the codomain is computable !

[1] assuming $\pi_{i,i+k}, \pi_{i,i+k} \neq 0$



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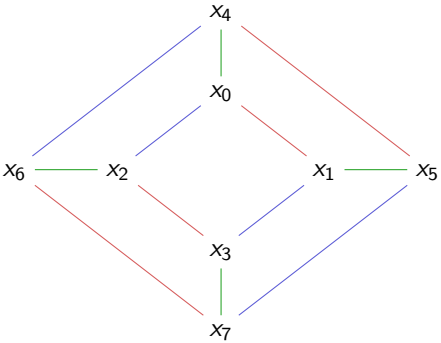
Formulae from graph theory

Let $\mathcal{T}$ be a spanning tree of G.



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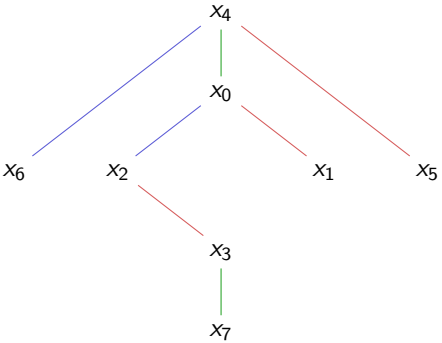

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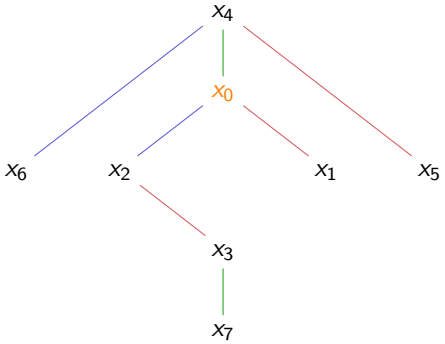

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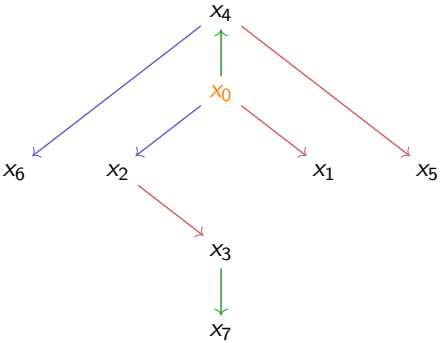

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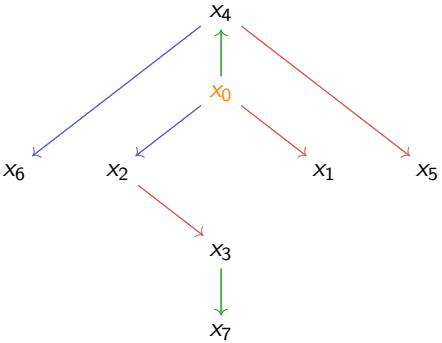

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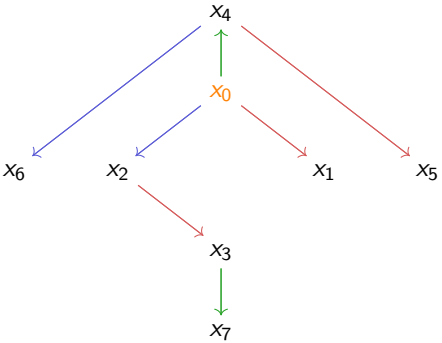
It is a valid solution HIIP !



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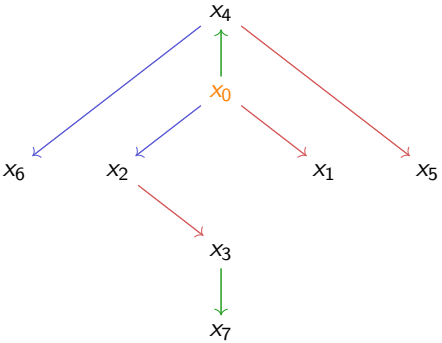

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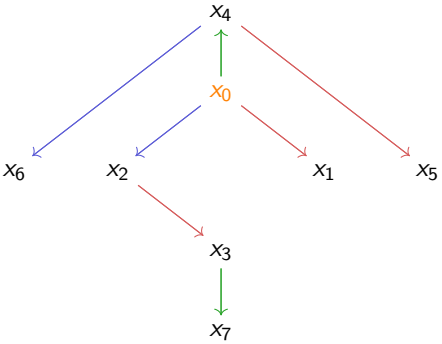
- Need good spanning tree of Hypercubes.



The case for Hamiltonian paths

- Introducing: $\mathcal{G}$ Gray paths on Hypercubes H_g.

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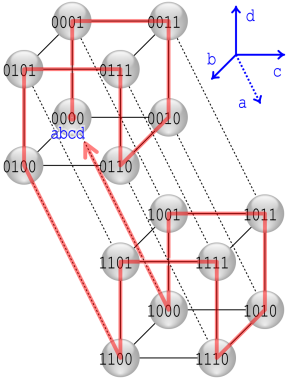

Figure: Gray code in a Tesseract

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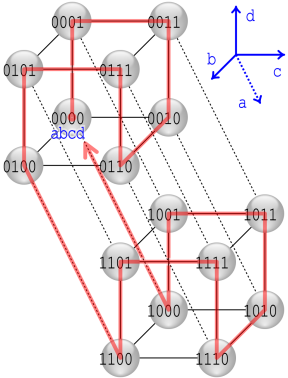
with $s_i := \eta(i+1) - \eta(i)$.



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Codomain computation in its full glory

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Figure: Structure of the products in an Hamiltonian path

- Computing HIIP can be done in $(3 \cdot 2^g - 6)\mathbf{M}$!
- Can be highly parallelised !

Retrospectively trivial

Algorithm 1: Solution to the HIIP

Data: HIIP input data, including a subset $S \subseteq (\mathbb{Z}/2\mathbb{Z})^g$, a graph $G := ((\mathbb{Z}/2\mathbb{Z})^g, E)$ and matrix $(\pi_{\mathbf{i},\mathbf{j}})_{\mathbf{i}\in(\mathbb{Z}/2\mathbb{Z})^g,\mathbf{j}\in S}$. A Hamiltonian path $\eta : [\![0, 2^g - 1]\!] \xrightarrow{\sim} (\mathbb{Z}/2\mathbb{Z})^g$ in G.

Result: A solution $(x_{\eta(i)}^{\odot -1})_{0\leq i\leq 2^g-1}$ to the HIIP defined by the input data (up to reordering by η).

$\rho_L(0) \leftarrow 1$;
$\rho_L(1) \leftarrow \pi_{\eta(1),\mathbf{s}_1}$;
$\rho_R(2^g - 1) \leftarrow 1$;
$\rho_R(2^g - 2) \leftarrow \pi_{\eta(2^g-1),\mathbf{s}_{2^g-2}}$;
for $i \in [\![2, 2^g - 1]\!]$ **do**
  $\rho_L(i) \leftarrow \rho_L(i-1) \cdot \pi_{\eta(i),\mathbf{s}_i}$;
  $\rho_R(2^g - 1 - i) \leftarrow \rho_L(2^g - i) \cdot \pi_{\eta(2^g-i),\mathbf{s}_{2^g-1-i}}$;
end
$x_{\eta(0)}^{-1} \leftarrow \rho_R(0)$;
$x_{\eta(2^g-1)}^{-1} \leftarrow \rho_L(2^g - 1)$;
for $i \in [\![1, 2^g - 2]\!]$ **do**
  $x_{\eta(i)}^{-1} \leftarrow \rho_L(i) \cdot \rho_R(i)$;
end
return $\left(x_{\eta(i)}^{-1}\right)_{0\leq i\leq 2^g-1}$; // Total cost: $(3 \cdot 2^g - 6)\mathbf{M}$

Generic isogeny: computing cost

	dim. g	dim. 4
Old [Dar25]	$(6 \cdot 2^g - 9)\mathbf{M} + g2^g\mathbf{S} + g^2 2^g\mathbf{a}$	$87\mathbf{M} + 64\mathbf{S} + 256\mathbf{a}$
New [DD26]	$3(2^g - 2)\mathbf{M} + g2^g\mathbf{S} + g^2 2^g\mathbf{a}$	$42\mathbf{M} + 64\mathbf{S} + 256\mathbf{a}$

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Graph theory gives us formulae that are:

- Simple to implement !
- Constant time !
- *Very flexible !*

Table of Contents

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   - Theta structures
   - Symmetric elements
   - Isogeny Formula
2. Computing isogenies using graphs
   - Isogeny computing
   - HIIP
3. Into the rabbit hole
   - Gluing
   - Special isogenies

Our arch nemesis: Gluing

$$\underbrace{(\prod^{g} E_i = A_0) \xrightarrow{f_1} A_1 \xrightarrow{f_2} A_2}_{\textit{Gluing}} \underbrace{\xrightarrow{f_2} \cdots \xrightarrow{f^{e-1}} A_{e-1} \xrightarrow{f_e} A_e}_{\text{generic}}$$

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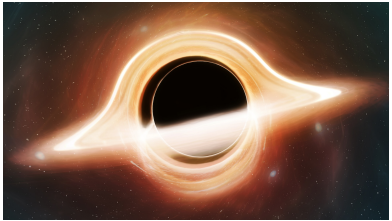

Figure: Artist view of a gluing isogeny

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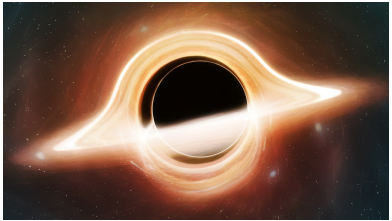
Gluing is:

- A rabbit-hole of technicality.
- HD-chain is 90% gluing (workload).



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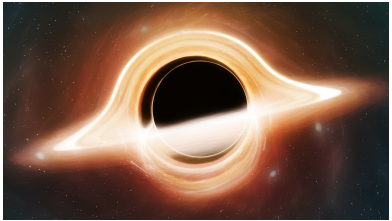
▶ HIIP is black-hole compatible !



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Dealing with zeros

$$\textbf{Problem with gluing}: \widetilde{\theta}_i^A(0) = 0 \text{ or } \widetilde{\theta}_i^A(f(T_j)) = 0 \implies G \subset H_g$$

[0] with $v \in \mathbb{Z}_2^g$ and $i \circ \sigma := (i_{\sigma(m)})_{1 \leq m \leq g}, \sigma \in \mathfrak{S}_g$

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Case 2: G is not connected.

Likely working with a product variety.

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such that $\Delta_{\sigma,v}(\mathcal{G}) \subset G$.

Figure: HD isogeny computing tool

Case 2: G is not connected.

Likely working with a product variety.

Solution: Use more points $T_k \implies$ More edges in G.

[0] with $v \in \mathbb{Z}_2^g$ and $i \circ \sigma := (i_{\sigma(m)})_{1 \leq m \leq g}, \sigma \in \mathfrak{S}_g$

Rabbits

In qt-Pegasis, gluing graphs G are rabbits !

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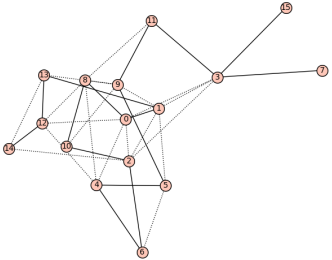

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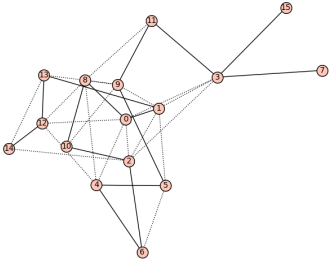

Figure: Rabbit picture

- Computing a rabbits take $(3 \cdot 2^g - 5)\mathbf{M}$
- Using other (many) technical tricks, made constant time gluing for qt-Pegasis.

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$$\text{Solving the HIIP over } G \iff \text{Solving the HIIP over } G_{/\overset{\sigma}{\sim}}$$

Example: Isogenies between fourfold of Weil-type $(2, 2)$

We have $\iota \in \mathrm{Aut}(A), \iota^2 = -1$

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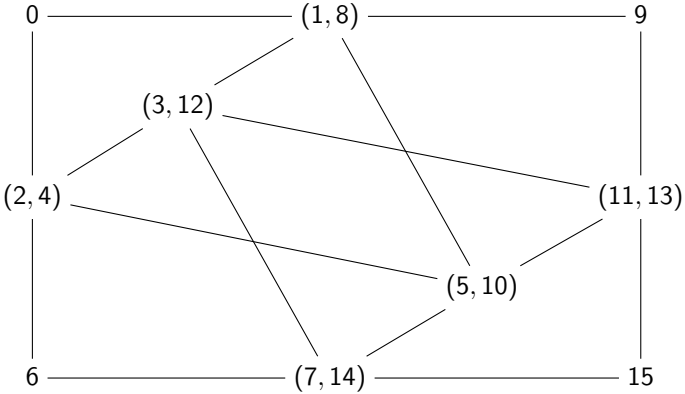

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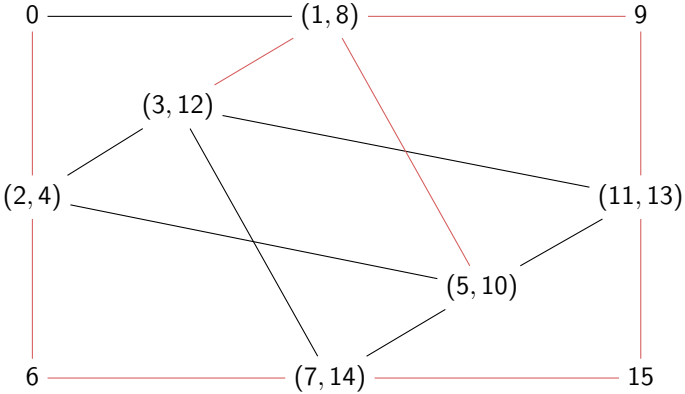

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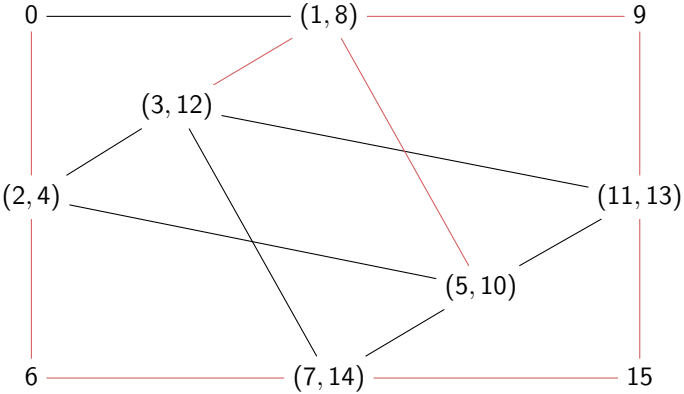

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Corollary

$\otimes$-MIKE is fast !

Implementation of qt-Pegasis

$\lceil \log_2(p) \rceil$	p	ARM (Mcycles)	X86 (Mcycles)	Timing (ms)
505	$27 \cdot 2^{500} - 1$	146.95	274.38	23.94
1008	$15 \cdot 2^{1004} - 1$	871.34	1627.05	146.19
1525	$5 \cdot 2^{1522} - 1$	3163.05	5208.81	543.01
2017	$5 \cdot 2^{2014} - 1$	7041.82	11192.45	1222.96
4030	$45 \cdot 2^{4024} - 1$	55228.89	83757.68	10654.93

Table: Performance of the C-implementation of 4-dimensional chain of qt-Pegasis in CPU Mcycles. Results are the average of 100 benchmark runs.

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1. Dim 4 chains are constructively practical!
2. We have all the tools for generic dimension!
3. It has never been easier to join the train!
4. Gluing will provide you *endless* fun!

Thanks for listening !

Beware of rabbits !

Ad break

Workshop on Abelian varieties and Applications to Post-Quantum Cryptography

16 - 20 November 2026
Bernoulli center, EPFL

More info on: https://bernoulli-isogeny.github.io/index.html

Figure: EPFL (not in November)

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